Ceren Sahin

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EDUCATION

University of Toronto, St. George Campus Sep 2023 – present
Bachelor of Science (BSc), Mathematics Specialist (Honors in the US), Computer Science Minor, expected 2027
Cumulative GPA: 3.89, Math GPA: 3.94, Math GPA percentage: 90.1 %

RESEARCH INTERESTS

Representation Theory, Algebraic Number Theory, p-adic Lie Theory

RESEARCH EXPERIENCE / READING COURSES

Reading Course – Representations of $SL_n(\mathbb{Z})$ for $n > 3$ May 2026 - Aug 2026
Supervisor: Prof. Daniel Litt, University of Toronto

- Read first nine chapters of *Analytic Pro-p Groups, Dixon et. al* and typed detailed exercise solutions.
- Wrote an expository paper on the proof of superrigidity principle for $SL_n(\mathbb{Z})$, adapting the argument by Bass, Milnor and Serre to this special case.

Research Project- Representation Theory May 2025 - Dec 2025
Supervisor: Prof. Mathilde Gerbelli-Gauthier, University of Toronto

- Established a bijective correspondence between two different parametrisations of cohomological representations of the unitary group $U(p,q)$, developed independently by M. Gerbelli-Gauthier and N. Bergeron.
- Co-authored a research manuscript detailing the construction and proof of the bijection.
- Studied the first 14 chapters of *Fulton & Harris,Representation Theory: A First Course* during the first two months of the meetings.
- This summer project continued as a reading course on group cohomology during Fall 2025 academic season.

Research Project - Analytic Number Theory May 2026 - Aug 2026
Supervisor: Prof. Asif Zaman, University of Toronto

- Studied the first ten chapters of Montgomery and Vaughan's *Multiplicative Number Theory*.
- Studied Backlund's method for investigating zeros of the Riemann zeta function through Edwards' *Riemann's Zeta Function*, and worked through proofs of explicit formulas relating zeros of Dirichlet L-functions to weighted sums over prime powers.
- Investigated explicit conductor bounds guaranteeing the existence of low-lying zeroes of primitive Dirichlet L-functions.

Reading Course – Algebraic Combinatorics & Lie Algebras Jan 2025 - Apr 2025
Supervisor: Prof. Hunter Spink, University of Toronto

- Read the first five chapters of *Algebraic Combinatorics* by Richard Stanley and the first eight chapters of *Introduction to Lie Algebras* by Karin Erdmann.
- Wrote comprehensive notes on finite dimensional representations of $\mathrm{sl}(2,\mathbb{C})$ and explored vector fields on $\mathbb{R}^2$ generated by exponentiation and Möbius transformations.

GRADUATE COURSEWORK

CORE COURSES

- Real Analysis I
- Real Analysis II
- Algebra I
- Algebra II
- Complex Analysis
- Algebraic Topology

ELECTIVE COURSES

- Algebraic Geometry (in progress)
- Algebraic Number Theory (in progress)
- Representation Theory
- Riemannian Geometry [final paper on the classification of locally symmetric Riemannian spaces]

AWARDS AND FUNDING

- University of Toronto Math Department Research Award - Undergraduate research funding 7,000 CAD (2026)
- University of Toronto Excellence Award — Undergraduate research funding, 7,000 CAD (2025)
- University of Toronto Scholars Award — Academic merit, 100,000 CAD (2023)

TEACHING EXPERIENCE

- MAT223 (Linear Algebra 1) Teaching Assistant. Led tutorials and graded assignments. Fall 2024 - present
- UT3-certified Student Tutor, providing tutoring for all first-year mathematics courses at UofT.